ECE101L Lab2

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May 2026

1 Thevenin and Norton Equivalents

1.1 Introduction

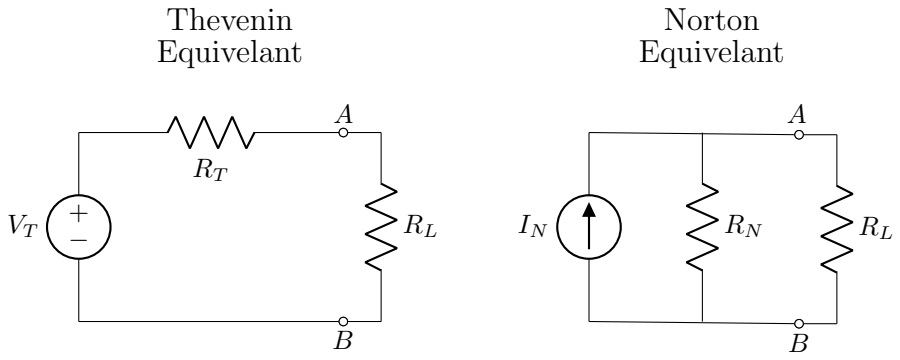


Figure 1: Thevenin and Norton Equivalents

Given any linear circuit with an input and output terminal, we can represent the internals as a source and a resistor. We can do this without ever knowing what is actually inside the circuit; all we need is access to those two terminals. We can also represent the load across those terminals as just a resistance.

There are two variations on this idea: Norton's Theorem, which represents the circuit with a current source and series resistor, and Thevenin's Theorem, which represents it as a voltage source and series resistor. Both are depicted in Figure 1. In theory, you should be able to replace your circuit with a Thevenin or Norton circuit and get the same behavior

The goal in this section is to find the two equivalent circuits for a simple linear circuit and prove that they can be used interchangeably. I predict that when we use either type of equivalent circuit to represent our original circuit, they will be nearly indistinguishable, which will be equivalent for our purposes.

1.2 Experiment

1.2.1 Measuring Norton and Thevenin Values

The circuit we will use to test the theorem on is a simple T-network with three resistors and a $5V$ source. Although we know what the circuit is, we need to prove we can find the equivalents with only the output terminals. Because of that, we will treat the circuit as completely unknown and only take measurements at nodes A and B. The difference between these two circuits is drawn in Figure 2.

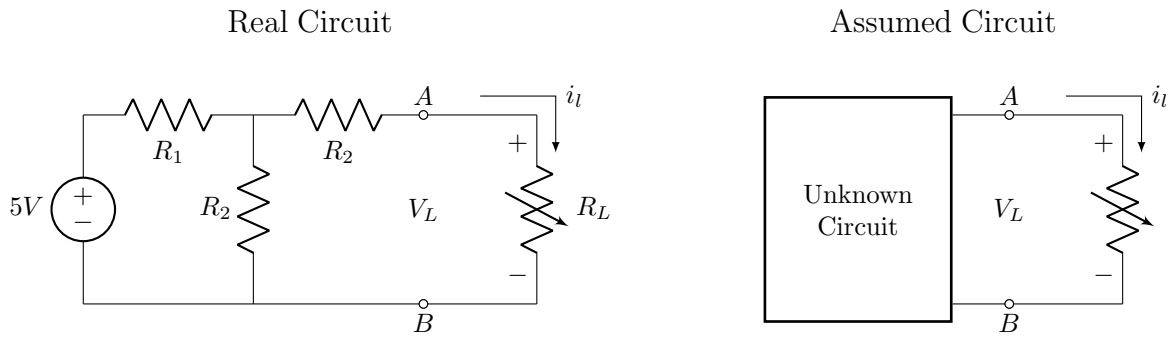


Figure 2: Real and Assumed Circuits

To find the equivalent values, we will measure the current and voltage across the terminals while the power source is connected, and we will measure the resistance while the value of the source is zero. To prove our circuits' equivalence, we will calculate Norton from Thevenin and Thevenin from Norton. Doing this will prove that the equivalence is true.

We should anticipate that we can convert the circuits between each other freely.

The voltage source value in a Thevenin circuit is the maximum voltage that can be supplied across the circuit terminals. The maximum voltage a circuit can supply to the terminals should be when the resistance is infinite. We can quantify this with Ohm's Law as $\lim_{R\to\infty} V_{Max} = R \cdot I$. Infinite resistance is an open circuit. We can do something similar for the maximum current across the terminals. The source value for Norton circuits is the maximum current the terminals can supply, which is $\lim_{R\to 0} I_{Max} = \frac{V}{R}$. Zero resistance is a short circuit.

To find the equivalent resistance, we need to set the value of all sources to zero, so that only the passive elements remain. Current sources become open segments, since there is zero current with infinite resistance, $\lim_{R\to\infty} I_{min} = \frac{V}{R}$. Voltage sources become shorts since there is no voltage change across ideal wires, $\lim_{R\to 0} V_{min} = R \cdot I$. This means we will replace our circuit source with a short. However, since we are supposed to treat the circuit as a black box, we need a second method that respects this. This will simply be an ohms law calculation, $R_{eq} = \frac{V_{Max}}{I_{max}}$. The equivalent resistance will be the same in Norton and Thevenin form.

Measuring the voltage across open terminals gives us the maximum voltage, $2.50V$, and the current across shorted terminals gives the maximum current, $16.81mA$. Altering the circuit to get our equivalent resistance gives 149.84Ω, which is very close to the calculated value from the current and voltage, 148.72Ω. Figure 3 is an updated version of Figure 1; It also has R_L removed for readability.

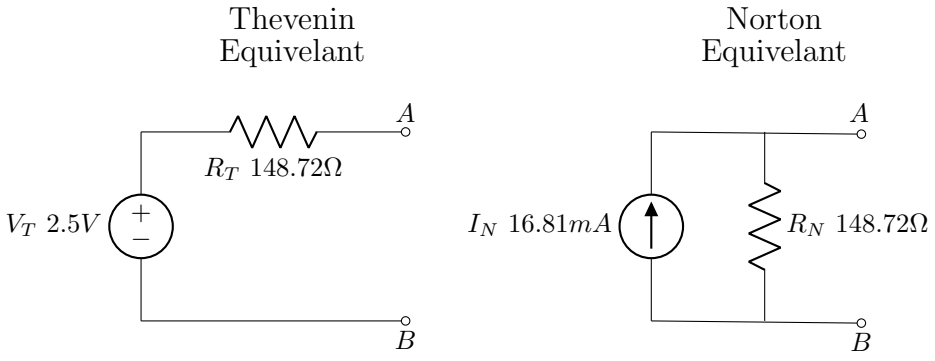


Figure 3: Thevenin and Norton Equivalents with Values

To convert them into each other, we can start by finding R_{eq} for each. For Thevenin to Norton, the voltage source becomes a short, so the circuit is reduced to only the resistor, so $R_T = R_N$. For Norton to Thevenin, we turn

the current source into an open segment; again, the circuit is reduced to the resistor only, so $R_N = R_T$.

To find I_N in the Thevenin to Norton case, we need to short the terminals. Doing this makes the circuit into a closed loop, so we can simply use Ohm's Law, $I_N = I_{Max} = \frac{V_T}{R_t}$, which is $16.81mA$. So our converted I_N is equal to the original I_N.

To find V_T from the Norton to Thevenin case, we need the voltage across the open terminals. Since the opening is in parallel to the resistor, the voltage across each is equal. We can again use Ohm's Law, $V_t = V_{max} = R_N \cdot I_N$, which is $2.5V$. Now our converted V_T is also equal to its original.

Since all our values can be found from either circuit, it means we can convert smoothly between them, which proves mathematically that these circuits are equivalent. We also verified that our two ways of finding the equivalent resistance, by altering the circuit and by calculating it, are approximately equal.

1.2.2 Comparing Models to Measurements

To further prove the equivalence of the two circuit types, we will use experimental data and prove that either circuit could have caused the recorded results. To do this, we will use various values for R_L and record the voltage and current across it. Previously, we only used infinite and zero resistance since they were needed to obtain the values V_T and I_N. If we can prove that the equivalent circuits are equal to our real one, it will also prove that they can predict the values seen on R_L due to the original circuit.

We can predict that the equivalence will be upheld when R_L is or is not an extreme value, that our readings can be explained by either Norton or Thevenin circuits, and that our equivalent circuits can be used to find the values on R_L in the real circuit.

Over seven trials, the following data in Table 1 was collected.

Table 1: Current and Voltage of R_L

Trial	1	2	3	4	5	6	7
V_{load}	$\approx 0V$	$1.26V$	$1.42V$	$1.77V$	$1.93V$	$2.05V$	$2.50V$
I_{load}	$16.81mA$	$8.16mA$	$7.14mA$	$4.86mA$	$3.77mA$	$2.99mA$	$\approx 0A$
R_{load}	$\approx 0\Omega$	152.90Ω	198.13Ω	369.70Ω	510.92Ω	683.75Ω	$\approx \infty\Omega$

Firstly, let's find some quick equations to determine V_{load} and I_{load} from either a Thevenin or a Norton circuit. For the current by the Thevenin circuit, since the resistors are in series, they will have the same current going through them. This means we can simply use Ohm's Law with the total circuit resistance.

$$I_{load\ by\ T} = \frac{V_T}{R_T + R_L}$$

For the voltage by a Thevenin circuit, it's easiest to use the voltage divider equation. It is valid here since we have two resistors in series and only one voltage source.

$$V_{load\ by\ T} = \frac{R_L}{R_L + R_T} \cdot V_T$$

To find the voltage by the Norton circuit, we can just use Ohm's Law again. Since the resistors are in parallel, the voltage across each is the same, so we can use Ohm's law with the total resistance.

$$V_{load\ by\ N} = \left(\frac{1}{R_L} + \frac{1}{R_N}\right)^{-1} \cdot I_N$$

Finlay, to find the current due to the Norton circuit, we can use the current divider equation. Since we have two parallel resistors and only one source its valid to use.

$$I_{load\ by\ N} = \frac{R_N}{R_L + R_N} \cdot I_N$$

Now, using these formulas, we can add on to Table 1. Table 2 compares the original trial data to the equation data.

Table 2: Compared R_L Data

Trial	1	2	3	4	5	6	7
V_{load}	≈ 0	$1.26V$	$1.42V$	$1.77V$	$1.93V$	$2.05V$	$2.50V$
I_{load}	$16.81mA$	$8.16mA$	$7.14mA$	$4.86mA$	$3.77mA$	$2.99mA$	≈ 0
R_{load}	$\approx 0\Omega$	152.90Ω	198.13Ω	369.70Ω	510.92Ω	683.75Ω	$\approx \infty\Omega$

$V_{load\ by\ T}$	$0V$	$1.26V$	$1.42V$	$1.78V$	$1.93V$	$2.05V$	$2.50V$
$I_{load\ by\ T}$	$16.81mA$	$8.28mA$	$7.20mA$	$4.82mA$	$3.78mA$	$3.00mA$	$0A$

$V_{load\ by\ N}$	$0V$	$1.26V$	$1.42V$	$1.78V$	$1.93V$	$2.05V$	$0A$
$I_{load\ by\ N}$	$16.81mA$	$8.28mA$	$7.20mA$	$4.82mA$	$3.78mA$	$3.00mA$	$2.50V$

Reading the table, it appears that all the values match and can be properly explained equally by either Thevenin or Norton circuits. However, it's important to note there is a small difference in currents. The Thevenin and Norton circuits have currents equal to each other, but they differ from the recorded values by around $0.05mA$ each. I believe the reason for this may be a small measurement error early on in the experiment. I believe it is due to electrical noise when measuring current. Since the currents are small, any motion during measurement can create electrical noise that can throw off readings, even very light vibrations. I have observed that these disturbances usually cause issues on scales smaller than a milliampere. Because of this, I believe such small differences can be ignored.

1.3 Conclusion

Using the data obtained, it was possible to find I_N, V_T, and R_T to define circuits that were equivalent to each other as well as the original circuit. Using simple circuit analysis, we were able to verify that the circuits are equivalent. Also, using simple circuit analysis methods, we were able to create equations to describe the values across a load resistance using these equivalent circuits. By showing how those equations could explain values seen on our real circuit, we proved they were equal to the real circuit and that our equivalent circuits can predict the values on a load caused by the real circuit.

2 Non-Ideal Power Sources

2.1 Introduction

In circuit analysis, power sources are assumed to have zero resistance and output exactly their stated values. In real circuits, however, Ideal circuit components, including ideal power sources, do not exist. Power sources will have an internal resistance (R_T or R_N) and will output values different from what they are expected to. Because of this, the power source used to run our experiment is presumably slightly off, and we can express this with a Thevenin circuit. If it is ideal, it will output exactly $5V$ and have zero internal resistance. I think we will be able to find a V_T and R_T to express the source as a Thevenin circuit, proving it is not an ideal source. Since we are Keysight's DC power supply, the NB1-class Benchtop DC Power Supply EDU36311A, we can expect the internal resistance to be close to 0.05Ω.

2.2 Experiment

For this experiment, we will move where our A and B nodes are, our voltage source will become the black box circuit, and the rest of the circuit will become our R_L. Figure 4 illustrates this small change.

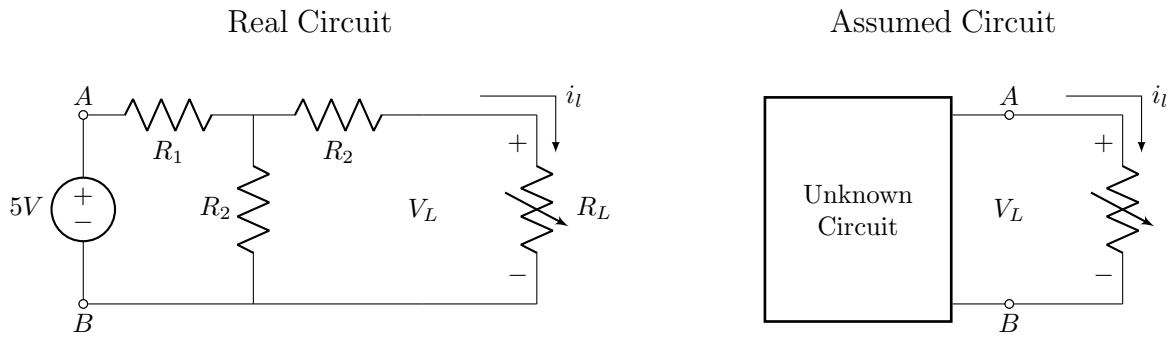


Figure 4: Ideal source setup

To begin, we will only need three measurements, V_{OC}, which is V_T, V_L, and I_L. After measuring, $V_T = 4.9994V$, $V_L = 4.9981V$ and $I_L = 24.315mA$. We need to use a higher level of precision than usual since we are expecting such a small resistance. With V_L and I_L, we can use Ohm's law to find the load resistance, which is 205.5562Ω. Now we can use a rearrangement of the equation for voltage by a Thevenin circuit from the earlier section:

$$V_{load\ by\ T} = \frac{R_L}{R_L + R_T} \cdot V_T$$

which becomes:

$$R_T = R_L(\frac{V_T}{V_L} - 1)$$

Using this tells us $R_T \approx 0.05346\Omega$. Now that we know what R_T and V_T are, we can draw our power supplies' equivalent circuit in Figure 5.

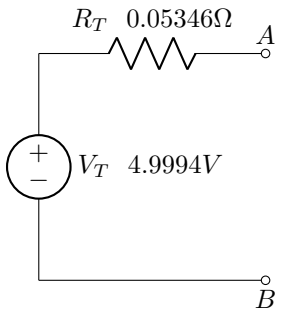


Figure 5: Actual Power Supply Circuit

2.3 Conclusion

Since the power supply is not exactly at its specified voltage, and since it has an internal resistance, that would mean it is a non-ideal power source. Non-ideal components become a larger issue for more complex or exact circuits. For most of our purposes, since the source is so close to the specified voltage, it can be considered ideal. It can be considered ideal for most applications. Bench supplies like this are designed to be as exact as possible, so a resistance less than a tenth of an ohm is a very good approximation.

3 Maximum Power Transfer

3.1 Introduction

As we can see from the data in Table 2, we have a maximum current and voltage. Our maximum current is $16.81mA$, and occurs when $R_L = 0\Omega$. Maximum voltage is $2.50V$ and occurs when $R_L \to \infty\Omega$. At both these points, the opposite value is zero. Due to the intermediate value theorem, we know there must be values between their maximums

and zero, when R_L is between 0 and ∞. Power is defined as $P = IV$. At the extreme values of R_L, since either current or voltage will be zero, the minimum power is zero. But what is the maximum power? We know that, due to the intermediate value theorem is must exist, since there are many points where neither voltage nor current is zero. The intent of this experiment is to find and prove what the maximum power, P_{Max}, is and what value of R_L causes it.

I predict that the maximum power will exist and will have a resistance that causes it between 0 and ∞. I think that when power is P_{Max}, $R_L = R_T$. I think this since the current and voltage dividers tell us that when resistances are equal, half the voltage will go through the resistor and half the current. If R_L deviates from that value, either voltage or current will be higher.

3.2 Experiment

Since power is defined as $P = IV$, we can calculate P with two of the equations from part one.

$$I_{load\ by\ T} = \frac{V_T}{R_T + R_L}$$

$$V_{load\ by\ T} = \frac{R_L}{R_L + R_T} \cdot V_T$$

I chose to use the equations made from the Thevenin circuit arbitrarily. Since it was proven that the Thevenin and Norton circuits are equivalent, it shouldn't matter. When you multiply them, you get the following equation.

$$P(R_L) = \frac{R_L \cdot V_T^2}{(R_L + R_T)^2}$$

V_T and R_T are constants found in part one. R_L is the variable since we want to find for what value of R_L $P = P_{Max}$. This also makes P a function of R_L. Below, in Figure 6, is a Graph of the equation. Six of the seven trials' powers were calculated and marked on the graph. I also marked where R_T will sit on the function since I theorized that it is where the maximum power is.

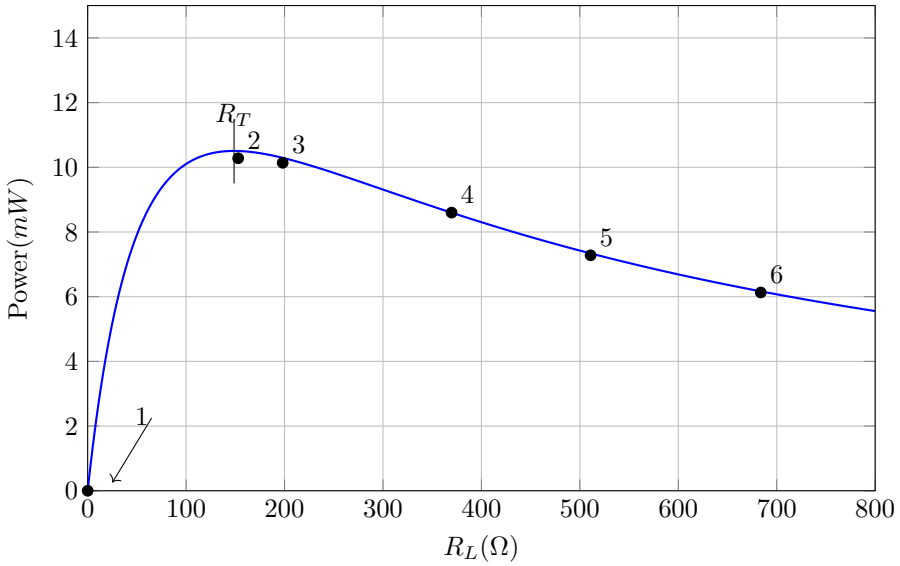


Figure 6: Power Provided at Load Resistance

Visually, it appears that R_T is the resistance for maximum power transfer. Trial 2 is also very close since its resistance is only off by 4.18Ω. Using the graph, we can safely say P_{Max} exists at a specific R_L in the domain $(0, \infty)$. Visual verification, however, is not a valid way to prove that $P(R_T) = P_{Max}$. We must prove this mathematically.

To find where the maximum is, we can differentiate the equation for $P(R_L)$ and solve for when it equals zero. Since peaks and valleys appear whenever the slope changes, this will let us find those points and classify them, including the peak in the graph. The largest peak will be the maximum.

$$\frac{dP(R_L)}{dR_L} = \frac{[(R_L + R_T)^2 - 2R_L(R_L + R_T)] \cdot V_T^2}{(R_L + R_T)^4} = \frac{R_T - R_L}{(R_T + R_L)^3} \cdot V_T$$

$$\frac{dP(R_L)}{dR_L} = \frac{R_T - R_L}{(R_T + R_L)^3} \cdot V_T = 0$$

The equality demonstrates that the original equation only has one peak at $R_L = R_T$. Negative values of R_L are not being considered since negative resistance isn't physically possible. This is another reason the domain of R_L is $(0, \infty)$, different from earlier, where we restricted R_L to be between resistance at V_{Max} and I_{Max}. Because of this restriction, the denominator will always be positive. At $R_L = R_T$ the derivative equals 0, we know this is a peak since for $R_L < R_T$, the slope is positive, and for $R_L > R_T$, the slope is negative. And since the derivative only has one point where it is zero, this peak is the maximum of the function, P_{Max}. It is therefore true that maximum power transfer occurs when $R_L = R_T$.

Knowing this relationship, we can create three simple equations for P_{Max} given the information about any Thevenin or Norton circuit. Firstly, using the equation for $P(R_L)$ where $R_L = R_T$, we can get:

$$P_{Max} = \frac{V_T^2}{4R_T}$$

We can also make a version for a Norton circuit since $V_T = I_N \cdot R_N$ and $R_T = R_N$. As well as a version that doesn't need R_T

$$P_{Max} = \frac{R_N \cdot I_N^2}{4}$$

$$P_{Max} = \frac{V_T \cdot I_N}{4}$$

3.3 Conclusion

Using the data and equations for our equivalent circuits, we were able to show what maximum power is, and when it happens. Trial two is so close to the maximum power, which was a coincidence, but it was a great example of maximum power. Maximum power transfer is a fundamental concept and is needed to build efficient devices. One example is radio frequency engineering, where you want to lose as little signal power as possible.